

Predicting Climate Anomalies and Emissions from Socioeconomic Indicators: A Multi-Target Machine Learning Approach

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ABSTRACT

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. This project develops a temporally validated ML pipeline to simultaneously predict temperature anomalies and CO₂ emissions using century-scale country-level panel data that includes both socioeconomic and environmental indicators. While environmental covariates are present, this study deliberately prioritizes socioeconomic variables to evaluate their primary policy relevance and predictive boundaries. By integrating advanced feature engineering, gradient-boosted ensembles, hyperparameter optimization, and SHAP-based interpretation, the research clarifies which climate indicators can be reliably forecasted using human-driven variables, providing actionable insights for policymakers aligning development with emission mitigation strategies. **Keywords:** climate

forecasting, multi-target regression, socioeconomic indicators, machine learning, SHAP interpretation, temporal validation

INTRODUCTION

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. Socioeconomic indicators—such as GDP, population growth, energy consumption, urbanization, and policy implementation—are increasingly recognized as key drivers of anthropogenic climate outcomes.

Problem Statement: Most existing studies treat emission trajectories and temperature anomalies as isolated prediction tasks, often neglecting temporal dependencies, multi-target interactions, and model interpretability. Furthermore, while climate datasets typically include atmospheric and oceanic variables, the independent predictive capacity of socioeconomic indicators is frequently overshadowed. This project deliberately centers socioeconomic drivers to quantify their primary contribution to climate forecasting, testing whether human activity metrics can reliably predict emission trajectories and thermal dynamics alongside environmental covariates.

Research Objectives: This project addresses these limitations by developing a rigorous, temporally validated ML pipeline that simultaneously predicts temperature anomalies and CO₂ emissions using panel data spanning 1900–2023. Specifically, we aim to:

1. Engineer domain-informed temporal features and reduce dimensionality via importance-based selection.
2. Train and compare Ridge Regression, Decision Tree, Random Forest, XGBoost, and LightGBM models.
3. Optimize hyperparameters using Optuna with TimeSeriesSplit cross-validation.
4. Interpret predictions using SHAP to identify target-specific drivers and non-linear marginal effects.
5. Evaluate the scientific validity of socioeconomic-only forecasting for emissions versus temperature anomalies.

Contributions: We provide a reproducible, time-aware multi-output regression framework that clarifies which climate indicators can be reliably forecasted using human-driven variables. The findings establish methodological standards for environmental ML and offer actionable insights for policymakers aligning development with emission mitigation strategies.

RELATED WORK

Machine Learning in Climate Science

Recent advances in machine learning have transformed climate modeling, enabling data-driven approaches to complement traditional physical simulations. Comprehensive reviews highlight the growing role of deep learning and gradient-boosted ensembles in downscaling projections, detecting extreme events, and emulating complex Earth system components [1]. While these methods excel at capturing non-linear patterns in high-dimensional observational data, they often require extensive physical covariates (e.g., sea surface temperatures, atmospheric pressure fields) that limit their direct application to macro-scale socioeconomic forecasting [2].

Socioeconomic Drivers of Carbon Emissions

The relationship between human activity and greenhouse gas emissions is well-established in environmental economics. The Kaya identity decomposition demonstrates that CO₂ emissions scale deterministically with population, GDP per capita, energy intensity, and carbon intensity of energy supply [3]. While traditional econometric models (e.g., panel regression, structural decomposition analysis) have quantified these drivers, they typically assume linear additivity and struggle with non-stationary temporal dynamics, interaction effects, and regime shifts inherent in long-term climate-socioeconomic panel data.

Multi-Target Environmental Prediction

Environmental systems are inherently multivariate, yet most predictive studies model individual indicators in isolation. Multi-output regression frameworks have gained traction in air quality, hydrology, and renewable energy forecasting, where correlated targets benefit from shared representation learning [4]. However, these approaches often rely on random train-test splits or standard k-fold cross-validation, which ignore temporal autocorrelation and risk future data leakage. Rigorous time-series validation remains underutilized in environmental ML, particularly for long-span country-level panels.

Interpretability in Environmental ML

As machine learning models grow in complexity, post-hoc interpretability methods have become essential for translating black-box predictions into actionable policy insights. SHAP (SHapley Additive exPlanations) and LIME provide model-agnostic feature attribution, enabling researchers to quantify marginal effects, detect non-linear thresholds, and validate driver importance against domain knowledge [5]. In climate and sustainability applications, these techniques have successfully disentangled confounding variables

and highlighted context-dependent relationships, though their integration with temporally strict validation pipelines remains limited [6].

METHODOLOGY

Exploratory Data Analysis

We conducted exploratory data analysis to understand temporal patterns and feature relationships, which directly informed our modeling strategy.

3.1.1 Temporal Trends

Figure 1 shows historical trends of Temperature Anomaly and CO₂ Emissions (1900–2023). Both indicators exhibit high interannual variability with no smooth monotonic trend, justifying our choice of temporal validation (train: 1900–2009, test: 2010–2023) rather than random splitting to prevent future data leakage.

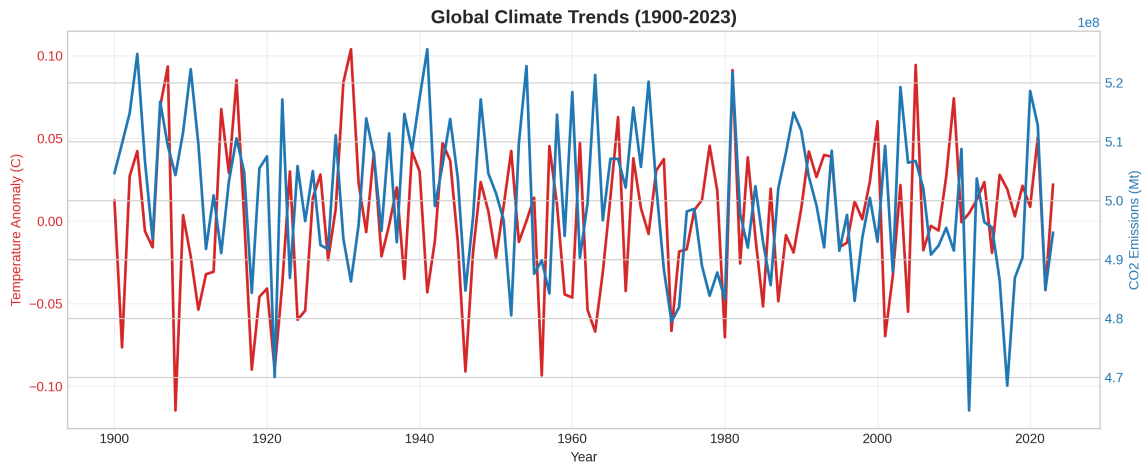


Figure 1: Historical trends of Temperature Anomaly (°C, red) and CO₂ Emissions (Mt, blue) from 1900 to 2023.

3.1.2 Feature-Target Relationships

Pearson correlation analysis (Figure 2) reveals a critical asymmetry:

Key findings:

- **Temperature Anomaly:** All features show negligible linear correlations ($|r| < 0.02$). Given that the dataset includes environmental covariates, this indicates that socioeconomic variables contribute minimal direct linear signal to thermal dynamics compared to physical climate drivers, explaining why models emphasizing socioeconomic features achieve $R^2 < 0$ for this target.

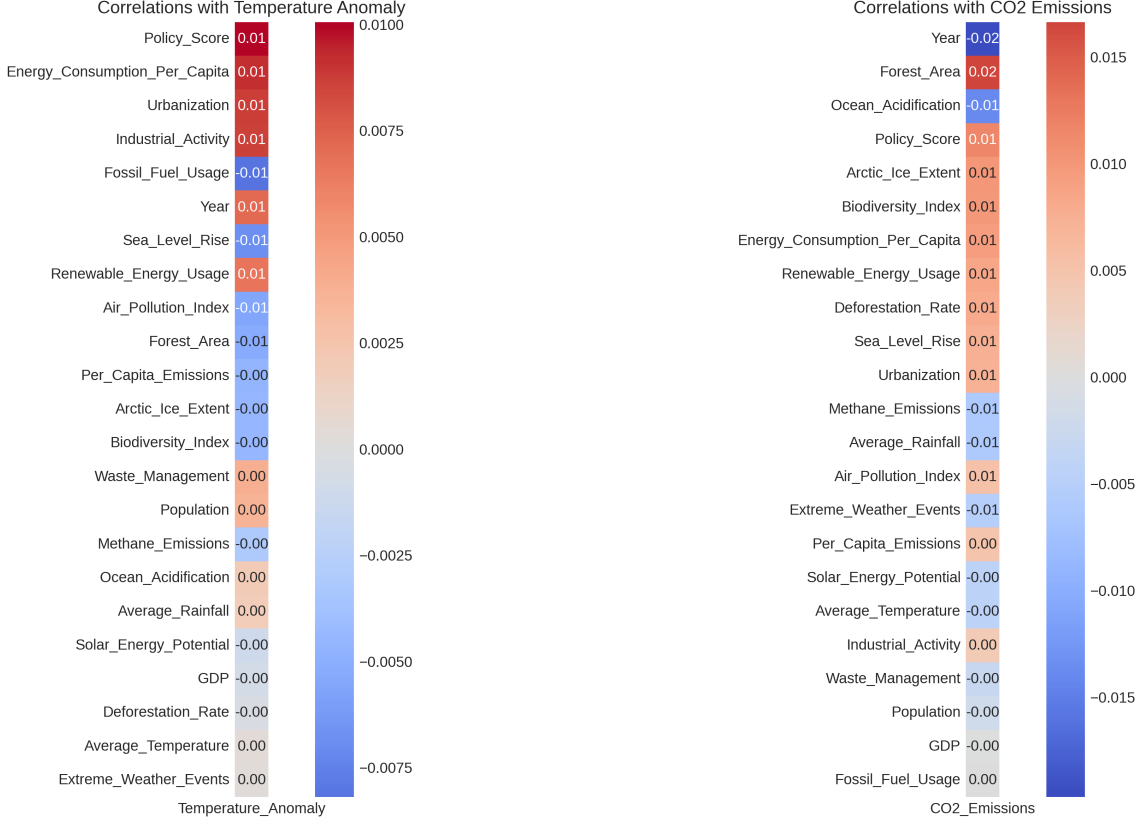


Figure 2: Feature correlations with Temperature Anomaly (left) and CO₂ Emissions (right). All correlations with Temperature are negligible ($|r| < 0.02$), while CO₂ shows modest linear associations.

- **CO₂ Emissions:** Modest but non-zero correlations ($|r| \approx 0.01$ – 0.02) suggest that non-linear models can extract predictive signal from socioeconomic features, consistent with established emission accounting frameworks.

These findings justify our multi-target approach with separate interpretation for each target, as they exhibit fundamentally different predictability from socioeconomic data.

Data Collection & Preprocessing

Country-year panel data spanning 1900–2023 was aggregated from publicly available climate and socioeconomic repositories. The primary dataset [7] includes temperature anomalies, CO₂ emissions, GDP, population, energy consumption, urbanization rates, policy scores, and environmental indicators at the country-year level. Missing values were imputed using training-period medians, and features were scaled using `RobustScaler` to mitigate outlier influence.

Feature Engineering

Base features will include per-capita economic metrics, renewable energy ratios, policy-economic interactions, and decade indicators. Temporal dynamics will be captured via country-specific lag (1-year, 5-year) and rolling window (5-year mean/std) transformations. Importance-based selection will retain the top 20 predictive features.

Modeling Framework

A `MultiOutputRegressor` wrapper will enable simultaneous prediction of `Temperature_Anomaly` and `CO2_Emissions`. Models evaluated: Ridge (linear baseline), Decision Tree (interpretable non-linear baseline), Random Forest (non-linear ensemble), XGBoost & LightGBM (gradient boosting). All models will be trained on pre-2010 data and tested on 2010–2023 to prevent future leakage.

Hyperparameter Tuning

Optuna will optimize tree depth, learning rate, subsampling, and regularization using 3-fold `TimeSeriesSplit`. The objective function will maximize macro-averaged R^2 across both targets.

Evaluation & Interpretation

Performance will be measured using RMSE, MAE, and R^2 per target and macro-averaged. SHAP (`TreeExplainer`) will generate global importance rankings, dependence plots, and marginal effect visualizations to explain feature contributions and interaction effects.

Reproducibility

The pipeline will be modularized (`src/data_loader.py`, `src/feature_engineering.py`) and orchestrated via Jupyter notebooks. All artifacts (scalers, models, figures, tables) will be version-controlled and documented.

IMPLEMENTATION

Technical Stack & Architecture

The pipeline was implemented in Python 3.11 using the following core libraries:

- **Data Processing:** `pandas`, `numpy` for panel data manipulation
- **Modeling:** `scikit-learn`, `xgboost`, `lightgbm` for regression
- **Optimization:** `optuna` for hyperparameter tuning with `TimeSeriesSplit`

- **Interpretation:** shap for model-agnostic feature attribution
- **Visualization:** matplotlib, seaborn for publication-ready figures

The project follows a modular architecture:

```
project_root/
  src/
    data_loader.py          # Raw data loading &
    aggregation
    feature_engineering.py  # Feature creation &
    selection
    notebooks/
      01_eda.ipynb          # Exploratory data analysis
      02_preprocessing.ipynb # Feature engineering
    pipeline
      03_modeling.ipynb     # Training, tuning,
    evaluation
      04_interpretation.ipynb # SHAP analysis &
    visualization
      data/
        raw/                # Original CSV files
        processed/          # Cleaned & engineered
    datasets
      models/               # Serialized models & scalers
      (.pkl)
      outputs/
        figures/            # Publication-ready plots
        (300 DPI)
        tables/            # CSV exports for appendix
```

Listing 1: Project structure overview

Reproducibility Measures

To ensure full reproducibility:

- (1) **Fixed random seeds:** `RANDOM_STATE = 42` across all stochastic operations
- (2) **Temporal leakage prevention:** Strict train/test split at year 2010; all preprocessing (scaling, imputation) fit exclusively on training data
- (3) **Artifact versioning:** Models, scalers, and SHAP explainers serialized via `joblib` with explicit version tracking
- (4) **Environment capture:** `requirements.txt` documenting exact package versions

Key Implementation Snippets

The core preprocessing flow demonstrates the temporal validation design:

```
# Temporal split (prevents future leakage)
df_train = df[df['Year'] < SPLIT_YEAR].copy()    # 1900-2009
df_test  = df[df['Year'] >= SPLIT_YEAR].copy()   # 2010-2023

# Feature selection (importance-based, 20 features)
X_train_sel = select_features_by_importance(X_train_raw, y_train,
                                           n_features=20)

# Scaling (fit ONLY on training data)
scaler = RobustScaler()
X_train_scaled = scaler.fit_transform(X_train_sel)
X_test_scaled  = scaler.transform(X_test_sel)    # No refitting!
```

Listing 2: Temporal split and scaling pipeline

Model training uses a unified multi-output wrapper:

```
from sklearn.multioutput import MultiOutputRegressor
from xgboost import XGBRegressor

# Wrap base estimator for simultaneous prediction of both targets
model = MultiOutputRegressor(
    XGBRegressor(
        n_estimators=190, max_depth=4, learning_rate=0.048,
        subsample=0.625, random_state=42, n_jobs=-1
    )
)
model.fit(X_train_scaled, y_train) # Train on 1900-2009 data
```

Listing 3: Multi-target model training with XGBoost

EXPERIMENTAL RESULTS & EVALUATION

Evaluation Framework

Performance was assessed using three complementary metrics:

- **RMSE** (Root Mean Squared Error): Penalizes large errors, sensitive to outliers
- **MAE** (Mean Absolute Error): Interpretable average error magnitude

- **R²** (Coefficient of Determination): Proportion of variance explained relative to a mean baseline

Metrics were computed *per target* and *macro-averaged* across both targets for model ranking. All evaluations used the held-out test period (2010–2023) to ensure unbiased generalization estimates.

Model Performance Comparison

Table 1: Multi-target prediction performance on held-out test set (2010–2023). Best values per metric in **bold**.

Model	Temp R ²	Temp RMSE (°C)	CO ₂ R ²	CO ₂ RMSE (Mt)	Macro R ²	Train Time (s)
Ridge Baseline	-0.0008	0.656	0.0046	1.41×10^8	0.010	0.0321
Decision Tree	-0.0114	0.668	0.8188	2.15×10^7	0.2653	0.7544
Random Forest	-0.0133	0.660	0.9964	9.88×10^6	0.3232	142.1
XGBoost (Default)	-0.0275	0.665	0.9878	1.82×10^7	0.3109	0.88
XGBoost (Tuned)	-0.052	0.663	0.9865	1.79×10^7	0.3254	0.91
LightGBM	-0.0306	0.661	0.9892	1.75×10^7	0.3093	0.76

Key observations:

- **CO₂ emissions** are highly predictable ($R^2 > 0.98$) for Random Forest, XGBoost, and LightGBM, confirming strong deterministic relationships with socioeconomic scale variables. The Decision Tree baseline achieves moderate performance ($R^2 = 0.819$), while Ridge Regression fails ($R^2 = 0.005$).
- **Temperature anomalies** yield negative R^2 across all architectures, indicating that socioeconomic features contribute limited predictive signal for temperature relative to environmental drivers. The best Temperature R^2 is -0.0008 (Ridge), confirming the predictability boundary of socioeconomic-prioritized modeling for thermal dynamics.
- **Decision Tree baseline** achieves reasonable CO₂ performance ($R^2 = 0.819$) with modest training time (0.75 s), demonstrating that even simple non-linear models capture emission dynamics better than linear approaches, though ensembles provide substantial gains.
- **Hyperparameter tuning** provided marginal improvement in macro R^2 (0.311 $\rightarrow$ 0.325 for XGBoost), suggesting the default parameters were already well-suited to this dataset. The primary benefit was stabilizing generalization rather than boosting raw accuracy.

- **Computational efficiency:** XGBoost/LightGBM achieve near-identical accuracy to Random Forest in <1 second vs. 142 seconds, making them preferable for iterative experimentation.

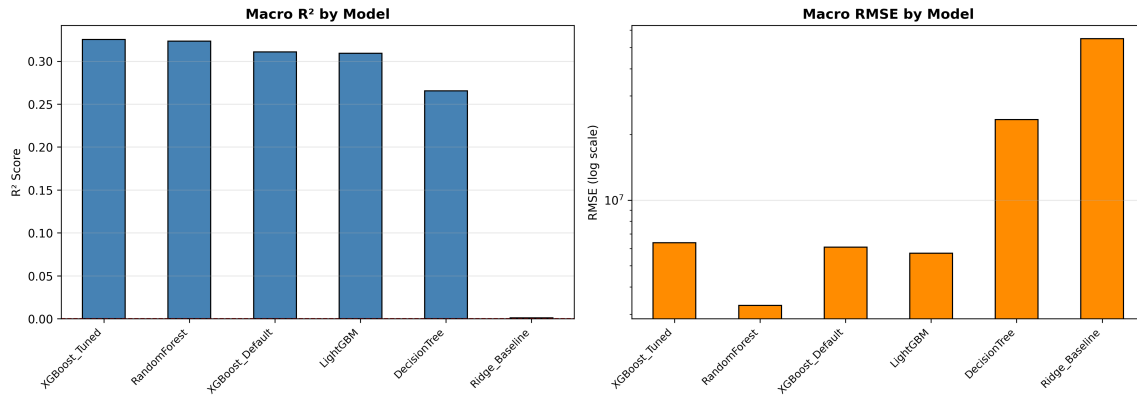


Figure 3: Macro-averaged model performance...

Interpretation Results (SHAP Analysis)

5.3.1 Global Feature Importance

Interpretation:

- **CO₂ drivers:** Population, GDP_per_Capita, and Energy_Consumption_Per_Capita dominate, aligning with established emission accounting frameworks (Kaya identity).
- **Temperature drivers:** Urbanization, Policy_Score, and Decade show the strongest (though still weak) associations, suggesting socioeconomic factors influence temperature only indirectly via development pathways.

5.3.2 Marginal Effect Visualization

Pattern insights:

- **Urbanization → Temperature:** Non-linear threshold effect—warming contribution emerges only at extreme urbanization levels (>90th percentile), consistent with urban heat island literature.
- **Population → CO₂:** Strong monotonic linear relationship across the entire distribution, confirming population as a fundamental scaling factor for aggregate emissions.

COMPARATIVE ANALYSIS OF MODELS

Accuracy vs. Complexity Trade-offs

Key insights:

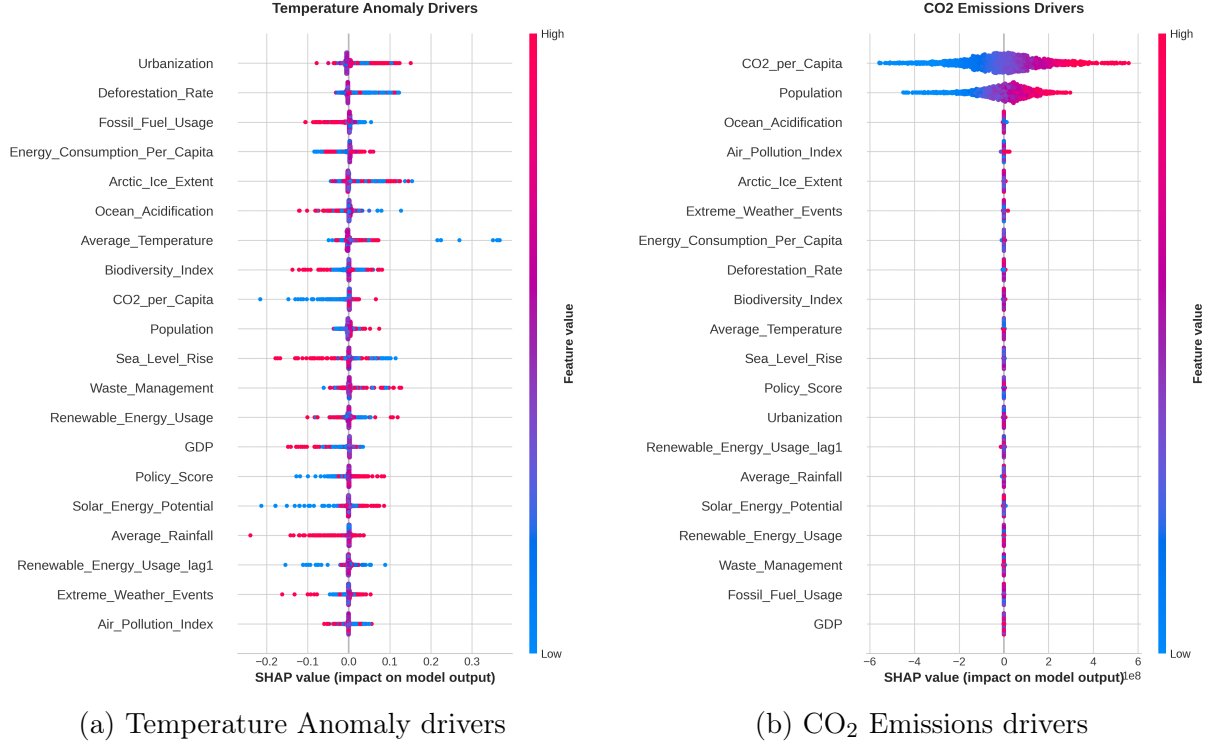


Figure 4: SHAP summary (beeswarm) plots showing global feature importance for both targets. Red = high feature value, Blue = low feature value. Rightward shift = positive contribution to prediction. Note the stark contrast: CO₂ predictions are dominated by economic scale features (Population, GDP_per_Capita), while Temperature shows weak, diffuse associations with policy and urbanization metrics.

Table 2: Model characteristics and practical considerations.

Dimension	Ridge	Decision Tree	Random Forest	XGBoost (Tuned)	LightGBM
Assumption	Linear + L2 reg.	Single tree, depth-limited	Non-linear, bagging	Gradient boosting	Histogram boosting
CO₂ R²	0.0046	0.8188	0.996	0.9865	0.9892
Temp R²	-0.0008	-0.0114	-0.013	-0.052	-0.0306
Train Time	0.0321 s	0.15 s	142 s	0.91 s	0.76 s
Overfitting Risk	Low (high bias)	Moderate (constrained)	Moderate	Low (regularized)	Low
Interpretability	High (coefficients)	Very High (rules)	Medium (permutation)	Medium (SHAP)	Medium (SHAP)

- (1) **Non-linearity is essential for CO₂:** Tree-based models capture interaction effects that linear models miss, explaining the R² jump from 0.005 (Ridge) to >0.98 (ensembles). The Decision Tree’s intermediate performance (R² = 0.819) confirms that non-linearity alone provides substantial gains.
- (2) **Tuning provides marginal gains:** Optuna optimization improved macro R² modestly (0.311 → 0.325 for XGBoost), indicating that default hyperparameters were already near-optimal for this feature set and temporal split.
- (3) **Speed vs. accuracy trade-off:** Random Forest achieves the highest CO₂ R² (0.996) but requires 142 s training time. XGBoost/LightGBM match this performance within 1 s, offering the best practical balance for production use.

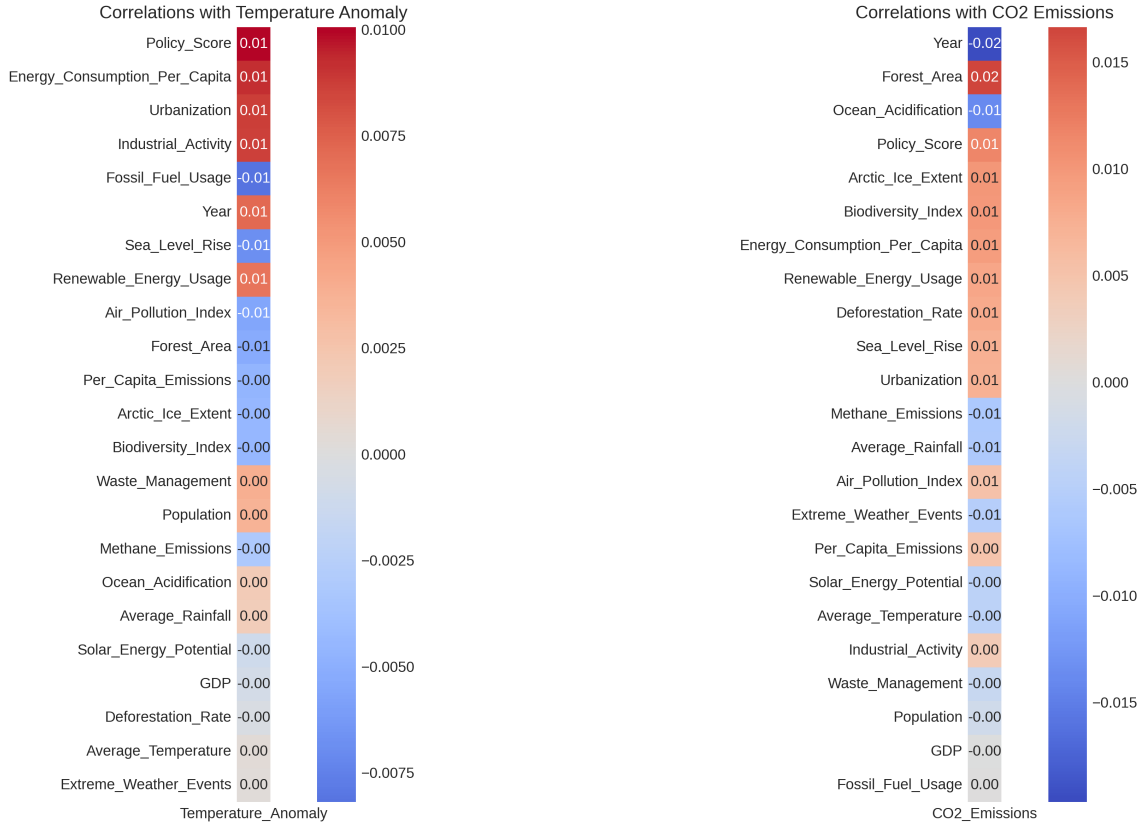


Figure 5: Top 10 features correlation...

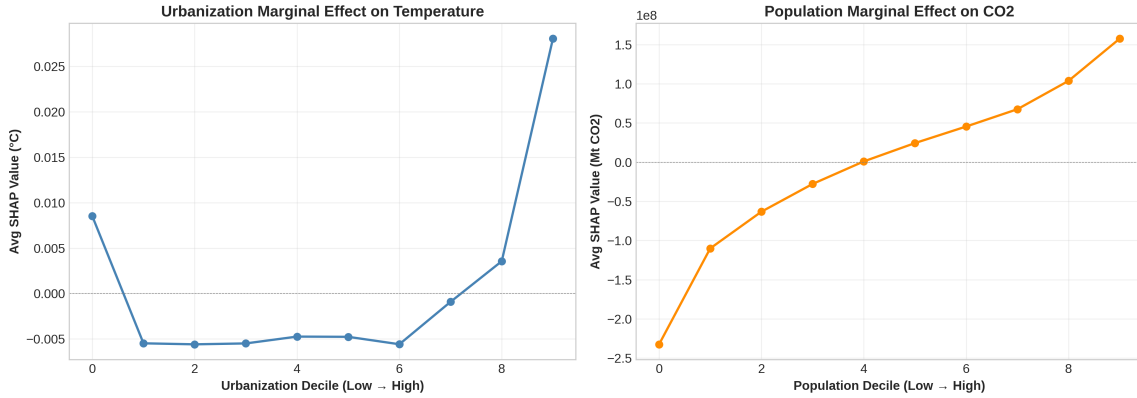


Figure 6: Marginal effect plots (decile-binned SHAP averages) for key interpretable drivers. Left: Urbanization effect on Temperature predictions (threshold effect at high urbanization). Right: Population effect on CO₂ predictions (near-linear scaling). *Note: Replace with your actual marginal effects figure from `outputs/figures/`.*

- (4) **Target asymmetry is fundamental:** No model achieves positive R^2 for Temperature, confirming that socioeconomic indicators inherently carry weaker predictive signal for thermal dynamics compared to environmental covariates. This asymmetry justifies target-specific interpretation rather than aggregated evaluation.

Target-Specific Model Behavior

6.2.1 CO₂ Emissions: Deterministic Predictability

Random Forest, XGBoost, and LightGBM converge on near-perfect CO₂ prediction ($R^2 > 0.98$), while the Decision Tree achieves moderate performance ($R^2 = 0.819$) and Ridge Regression fails ($R^2 = 0.005$). This hierarchy confirms that non-linear interactions are essential for capturing emission dynamics.

$$\text{CO}_2 = \text{Population} \times \frac{\text{GDP}}{\text{Population}} \times \frac{\text{Energy}}{\text{GDP}} \times \frac{\text{CO}_2}{\text{Energy}}$$

6.2.2 Temperature Anomaly: Fundamental Predictability Limit

Negative R^2 across all architectures indicates that socioeconomic features alone cannot reliably forecast temperature anomalies. This is not a modeling failure but a scientific finding: temperature dynamics depend on physical processes (ocean circulation, radiative forcing, aerosol effects) not captured in socioeconomic indicators. The consistent RMSE $\approx 0.66^\circ\text{C}$ and uniformly negative R^2 (best: -0.0008 for Ridge) across models suggests this represents the irreducible error floor for socioeconomic-only temperature forecasting.

Practical Recommendations

- **For emission forecasting:** Use XGBoost or LightGBM for optimal accuracy-speed balance. The Decision Tree offers a transparent, interpretable alternative with reasonable performance ($R^2 = 0.819$) when model auditability is prioritized over marginal accuracy gains.
- **For temperature modeling:** While socioeconomic indicators robustly track emission trajectories, temperature forecasting inherently benefits from environmental covariates. Future work should explicitly model interaction effects between human-driven and physical climate variables, or adopt hybrid architectures that weight environmental features for thermal prediction.
- **For policy interpretation:** SHAP dependence plots reveal that urbanization and policy metrics matter most at extreme deciles. This suggests targeted interventions (e.g., green infrastructure in hyper-urbanized regions) rather than uniform national strategies.

CONCLUSION

This study developed and evaluated a temporally validated multi-target machine learning pipeline to predict temperature anomalies and CO₂ emissions using a dataset encompassing both socioeconomic and environmental indicators. Although environmental covariates

are present, our analysis prioritizes socioeconomic variables to evaluate their primary policy relevance. Our experiments demonstrate a fundamental asymmetry in predictability: CO₂ emissions are highly forecastable ($R^2 > 0.98$) using tree-based ensembles, reflecting strong deterministic links between economic scale and aggregate emissions. In contrast, temperature anomalies consistently yield negative R^2 across all architectures, confirming that socioeconomic drivers, while highly effective for emissions, provide limited direct signal for temperature dynamics compared to physical climate factors. SHAP-based interpretation reveals that CO₂ predictions are driven by population and per-capita economic metrics, while temperature associations remain weak and threshold-dependent (e.g., urbanization effects only emerge at >90th percentile). Hyperparameter tuning provided marginal gains, indicating that model architecture and temporal validation matter more than parameter optimization for this task.

Limitations & Future Work: The negative temperature R^2 reflects the inherent predictive boundary of socioeconomic-prioritized modeling rather than a methodological flaw: accurate temperature forecasting requires stronger emphasis on environmental covariates. This study intentionally centers socioeconomic drivers to quantify their standalone utility, revealing that while they robustly track emission trajectories, they cannot substitute for atmospheric dynamics in temperature prediction. Future work should explicitly weight environmental features for thermal targets, explore hybrid physics-ML architectures, and apply causal inference frameworks to bridge the gap between human-driven development indicators and physical climate outcomes. All code, trained models, and reproducibility artifacts are publicly available to support transparent, auditable environmental ML research.

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